Republic of the Philippines
Western Mindanao State University
College of Computing Studies
DEPARTMENT OF INFORMATION TECHNOLOGY
Zamboanga City

VisionPool: A Real-Time Pool Rule Compliance Monitoring System Using Object Detection and Automated Flagging for Administrative Oversight

A CAPSTONE PROJECT

Presented to the Faculty of the
College of Computing Studies
Western Mindanao State University

In Partial Fulfillment
of the Requirements for the Degree
Bachelor of Science in Information Technology

SOLIS, LUISA MAE C.
Researcher

December, 2026

EXECUTIVE SUMMARY

VisionPool is a real-time pool rule compliance monitoring system designed to automate the detection and documentation of rule violations in public swimming pool facilities. The system leverages YOLOv8, a state-of-the-art object detection framework, integrated with IP camera feeds via OpenCV, to identify three primary violation types: absence of entry wristbands, improper swimming attire, and prohibited jumping or diving activities.

The system addresses a critical operational gap in public pool management, where manual enforcement of safety rules is often inconsistent, labor-intensive, and prone to human error, particularly during peak operational hours. By automating rule monitoring, VisionPool enables pool administrators and staff to respond to violations promptly and maintain accurate, timestamped incident documentation through a Django-based web administrative dashboard.

The project follows the Agile software development methodology to allow iterative refinement of the detection model and dashboard features based on continuous feedback. The detection pipeline captures violation screenshots, records timestamps and violation type labels, and transmits these records to the web dashboard in real time. The system is evaluated based on accuracy, precision, and recall metrics derived from a locally collected dataset of pool environment images.

The expected output is a functional, deployable pool compliance monitoring system that reduces the workload of pool management staff, improves rule enforcement consistency, and provides a reliable digital record of violations for administrative review and decision-making.

DEDICATION

This work is dedicated to the pool management staff and administrators who tirelessly ensure the safety and well-being of every patron in public recreational facilities. Your dedication to public service inspired this project.

To my family, whose unwavering support and encouragement made this undertaking possible.

And to the faculty of the Department of Information Technology of Western Mindanao State University, whose guidance shaped this research into its final form.

ACKNOWLEDGEMENT

The researcher extends her sincerest gratitude to all those who contributed to the successful completion of this capstone project.

To the Department of Information Technology faculty of Western Mindanao State University, for their invaluable guidance, constructive feedback, and unwavering support throughout the entire research process.

To the pool management staff and administrators of the target institution, for their cooperation during data collection, system validation, and user acceptance testing, and for generously sharing their operational knowledge and expertise.

To the members of the panel of evaluators, for their thorough and insightful comments that significantly improved the quality of this study.

To the open-source community behind YOLOv8, OpenCV, Django, and Roboflow, whose freely available tools and resources made the development of this system possible.

Finally, to family and friends, for their patience, encouragement, and moral support throughout this journey.

ABSTRACT

Public swimming pool facilities rely on consistent rule enforcement to ensure patron safety and operational integrity. However, manual monitoring of pool rules such as wristband verification, proper swimming attire, and restrictions on jumping or diving is often unreliable, especially during high-traffic periods. This capstone project presents VisionPool, a real-time pool rule compliance monitoring system that employs YOLOv8-based object detection to automatically identify rule violations from live IP camera feeds.

The system integrates OpenCV for real-time video stream processing, a fine-tuned YOLOv8 model for violation detection, an automated screenshot and flagging mechanism, and a Django-based web dashboard for administrative oversight. The detection pipeline identifies three violation categories: missing wristbands, improper swimming attire, and prohibited jumping or diving, and relays flagged incidents with timestamps to the administrative dashboard for review and record-keeping.

The project adopts the Agile development methodology to facilitate iterative model training and system refinement. The detection model is trained and evaluated on a locally collected dataset of pool environment images, annotated using Roboflow, and assessed based on precision, recall, and mean Average Precision (mAP) metrics. The system is designed to operate in real time on a local network using an affordable Tapo IP camera as the primary input device.

VisionPool aims to reduce the operational burden on pool management staff, improve consistency in rule enforcement, and provide a digital incident log that supports data-driven administrative decision-making. The system contributes to the broader field of computer vision applications in public safety and facility management.

Keywords: *object detection, YOLOv8, pool compliance monitoring, real-time video analysis, administrative dashboard, OpenCV, Django*

CHAPTER 1

INTRODUCTION

Project Context

Public swimming pools are among the most widely patronized recreational facilities, offering communities access to leisure, fitness, and competitive sports activities. In the Philippines, public and private pool facilities are governed by a combination of facility-specific rules and public health standards that aim to protect patrons and maintain operational order. Common pool policies include the mandatory use of entry wristbands to verify paid admission, requirements for proper swimming attire, and designated restrictions against jumping or diving in non-sanctioned areas. These rules are essential not only for ensuring fair access to the facility but also for preventing accidents and injuries.

Despite the importance of consistent rule enforcement, most pool facilities continue to rely on manual monitoring by lifeguards and pool attendants. This approach is inherently limited by human attention span, staffing levels, and the physical constraints of monitoring expansive pool areas during peak hours when patron density is highest. Violations frequently go unnoticed or are addressed inconsistently, leading to potential safety hazards and disputes between patrons and staff.

Recent advances in computer vision and deep learning have demonstrated significant potential for automating visual monitoring tasks in public spaces. YOLOv8 (You Only Look Once, version 8), developed by Ultralytics, is one of the most efficient and accurate real-time object detection frameworks available today. Its ability to process live video streams with minimal computational overhead makes it well-suited for deployment in resource-constrained environments such as local pool facilities. Paired with affordable IP cameras and web-based administrative tools, YOLOv8 enables the construction of cost-effective automated monitoring systems that can operate continuously without fatigue.

VisionPool is proposed as a direct response to the operational limitations of manual pool rule enforcement. By deploying a YOLOv8-based detection pipeline connected to a Tapo IP camera and a Django-powered administrative dashboard, VisionPool aims to provide pool management staff with a reliable, data-driven tool for real-time rule compliance monitoring, automated violation flagging, and incident documentation.

Statement of the Problem

Public swimming pool facilities face persistent challenges in the consistent and reliable enforcement of safety and operational rules. Specifically, the following problems have been identified:

1. Manual monitoring of pool rules is labor-intensive and subject to human error. Lifeguards and pool attendants cannot maintain constant vigilance over all patrons simultaneously, particularly during peak hours, resulting in undetected or inconsistently addressed violations.
2. The absence of a systematic violation documentation process makes it difficult for pool management to track compliance trends, identify repeat violators, or provide evidence-based justification for operational decisions or disciplinary actions.
3. Existing pool management systems do not integrate real-time visual monitoring capabilities, leaving a significant gap between administrative oversight tools and actual on-ground rule enforcement.
4. Manual rule enforcement places additional stress on pool staff and may lead to confrontational situations with patrons when violations are raised inconsistently.

These problems collectively reduce the effectiveness of pool safety protocols and increase the risk of patron injury, unauthorized access, and non-compliance with facility regulations. VisionPool addresses these issues by automating the detection and documentation of rule violations through computer vision technology.

Purpose and Description

VisionPool is a real-time pool rule compliance monitoring system designed to automate the identification and documentation of rule violations in public swimming pool facilities. The system uses a YOLOv8-based object detection model trained on pool environment images to detect three primary violation types from live IP camera feeds: the absence of entry wristbands, non-compliance with proper swimming attire requirements, and prohibited jumping or diving activities.

Upon detecting a violation, the system automatically captures a screenshot of the flagged incident, appends a timestamp and violation type label, and transmits the record to a Django-based web administrative dashboard accessible to pool management staff. The dashboard provides a real-time display of flagged incidents and maintains a searchable violation log for administrative review, reporting, and record-keeping purposes.

The system is designed for deployment in a local network environment using an affordable Tapo IP camera as the video input source and OpenCV as the video processing framework. VisionPool is intended to complement existing pool management operations rather than replace human staff, providing an additional layer of automated oversight that improves enforcement consistency and reduces the manual workload of pool attendants and administrators.

Objectives

General Objective: To design, develop, and evaluate a real-time pool rule compliance monitoring system that uses object detection to automatically flag violations and relay them to an administrative dashboard for oversight and documentation.

Specific Objectives

5. To develop a YOLOv8-based detection model capable of identifying wristbands, swimming attire compliance, and jumping or diving activities from live IP camera feeds.
6. To implement an automated flagging and screenshot capture mechanism that records violation incidents with timestamps and violation type labels.
7. To develop a web-based administrative dashboard using Django that displays flagged incidents in real time and maintains a searchable and filterable violation log.
8. To integrate the detection pipeline with a Tapo IP camera input via OpenCV for real-time video stream processing.
9. To evaluate the accuracy, precision, recall, and mean Average Precision (mAP) of the detection model on a locally collected and annotated dataset of pool environment images.

Scope and Limitation

VisionPool covers the following functionalities and operational boundaries:

The system detects three specific rule violation types: (1) absence of entry wristbands, (2) improper swimming attire, and (3) prohibited jumping or diving activities. Detection is performed in real time from a single Tapo IP camera feed processed via OpenCV.

The administrative dashboard, developed using the Django web framework, provides real-time display of flagged violations, a searchable violation log, and screenshot viewing capabilities. The system is intended for deployment on a local network accessible to authorized pool management staff.

The YOLOv8 model is trained and evaluated on a locally collected dataset of pool environment images. The system operates on a desktop or laptop computer connected to the same local network as the camera.

The following limitations apply to the current scope of the project: The system is designed for single-camera deployment and does not natively support multi-camera configurations in this version. Detection accuracy is dependent on camera placement, image resolution, and lighting conditions. The system does not perform facial recognition or patron identification. The violation detection model may require periodic retraining as pool environment conditions or rule definitions change. The system does not automatically trigger alerts to external agencies and is intended solely for internal administrative use.

Significance of Study

VisionPool addresses a practical and underserved need in the management of public swimming pool facilities. Its significance extends to the following stakeholders:

Pool Management Staff and Administrators: The primary beneficiaries of VisionPool are pool management personnel, including pool managers, supervisors, and administrative staff. The system reduces the manual burden of rule enforcement monitoring, provides a reliable and objective record of violations, and enables data-driven decision-making regarding pool operations, staffing, and patron compliance.

Lifeguards and Pool Attendants: By providing an automated violation detection layer, VisionPool allows lifeguards and pool attendants to focus their attention on patron safety and emergency response rather than constant rule monitoring. This reduces occupational stress and the risk of confrontational interactions with patrons over rule enforcement.

Pool Patrons: A consistently enforced rule environment benefits all pool patrons by ensuring fair application of facility regulations, reducing the risk of accidents caused by rule violations such as unauthorized jumping or improper attire, and maintaining the overall quality of the recreational experience.

The Field of Computer Vision and Public Safety: VisionPool contributes to the growing body of applied research on computer vision systems in public facility management. The project demonstrates the feasibility of deploying affordable, real-time object detection systems in resource-constrained environments and provides a replicable framework for similar monitoring applications in recreational and public safety contexts.

Western Mindanao State University: As a capstone research output, VisionPool contributes to the institutional knowledge base of the Department of Information Technology and demonstrates the practical application of advanced machine learning techniques in addressing real-world community needs.

Definition of Terms

The following terms are defined as used in this study:

Term	Definition
Administrative Dashboard	A web-based interface accessible to authorized pool management personnel that displays real-time flagged violations, violation logs, and incident screenshots captured by the VisionPool system.
Automated Flagging	The process by which the system automatically identifies and marks a detected rule violation for administrative review without requiring manual intervention from pool staff.
Bounding Box	A rectangular region drawn around a detected object in an image or video frame, used by the YOLOv8 model to indicate the location and extent of a detected person or object.
Deep Learning	A subset of machine learning that uses neural networks with multiple layers to learn complex patterns from data, enabling tasks such as image classification and object detection.
Django	An open-source, high-level Python web framework that enables rapid development of secure and maintainable web applications, used in this study to develop the administrative dashboard.
Fine-tuning	The process of taking a pre-trained machine learning model and further training it on a domain-specific dataset to improve its performance on a targeted task, applied in this study to adapt YOLOv8 to pool environment violation detection.
IP Camera	An Internet Protocol camera that transmits video data over a network, used in this study as the primary video input device for real-time pool area monitoring.
Mean Average Precision (mAP)	A standard evaluation metric for object detection models that measures the average precision across multiple object classes and detection

	thresholds, used to assess the overall accuracy of the VisionPool detection model.
Object Detection	A computer vision technique that identifies and localizes specific objects within an image or video frame by classifying detected regions and drawing bounding boxes around them.
OpenCV	Open Source Computer Vision Library; an open-source library of programming functions for real-time computer vision, used in this study to capture and process live video streams from the IP camera.
Precision	An evaluation metric representing the proportion of detected violations that are true violations, measuring the model's ability to avoid false positives.
Real-Time Processing	The ability of a system to process and respond to input data, such as live video frames, as it is received, with sufficiently low latency to support immediate decision-making.
Recall	An evaluation metric representing the proportion of actual violations that are correctly detected by the model, measuring the model's ability to avoid false negatives.
Roboflow	A cloud-based platform for managing, annotating, and preprocessing machine learning datasets, used in this study for image annotation and dataset preparation in YOLOv8 format.
RTSP (Real Time Streaming Protocol)	A network protocol used to control the delivery of real-time media streams from IP cameras, used in this study to access the video feed from the Tapo camera via OpenCV.
Screenshot Capture	The automated saving of a still image from the video stream at the moment a rule violation is detected, used to create a timestamped visual record of each flagged incident.
Wristband	A physical band worn on the wrist by pool patrons to indicate paid entry authorization, the presence or absence of which is monitored by the VisionPool system as a compliance indicator.
YOLOv8	You Only Look Once version 8; a state-of-the-art real-time object detection framework developed by Ultralytics, used in this study as the core machine learning model for detecting pool rule violations from live video feeds.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND SYSTEMS

Introduction

The development of VisionPool draws upon an expanding body of research spanning real-time object detection, computer vision-based compliance monitoring, human activity recognition, wearable and clothing attribute detection, automated surveillance systems, and web-based administrative interfaces for institutional monitoring. The following review examines twenty related works published between 2016 and 2026 that share thematic and technical relevance with the proposed study. The reviewed works are organized into five thematic clusters: (1) YOLO-based object detection frameworks; (2) computer vision for compliance and safety monitoring; (3) human activity and behavior recognition; (4) clothing and accessory attribute detection; and (5) web-based monitoring dashboards and administrative systems. Each entry discusses the study's contribution, methodology, key findings, and its relevance and limitations relative to VisionPool.

YOLO-Based Object Detection Frameworks

RRL 1: Development of YOLOv3 for Multi-Class Object Detection

Redmon and Farhadi (2018) presented YOLOv3, an improved iteration of the YOLO object detection framework that introduced multi-scale detection using feature pyramid networks, enabling more accurate detection of objects at varying sizes within a single image. The study demonstrated state-of-the-art detection speed and competitive accuracy on the COCO benchmark dataset, establishing YOLOv3 as one of the most widely adopted frameworks for real-time detection tasks. This work is foundational to VisionPool in that it established the multi-scale feature extraction principles that were refined in subsequent YOLO versions, including YOLOv8. The limitation of YOLOv3 relative to the present study is its lower detection accuracy on small objects compared to later architectures, which is a critical consideration for wristband detection in pool environments.

RRL 2: YOLOv4 — Optimal Speed and Accuracy of Object Detection

Bochkovskiy et al. (2020) introduced YOLOv4, integrating several architectural improvements including CSPDarknet53 as the backbone, SPP (Spatial Pyramid Pooling), and PANet path aggregation, resulting in significantly improved detection accuracy while maintaining real-time inference speeds on commodity GPUs. The study confirmed that high-performance object detection could be achieved without dedicated high-end computing infrastructure, making YOLO-family models practical for deployment in institutional settings. The relevance to VisionPool lies in the demonstration that YOLO architectures can be fine-tuned for domain-specific datasets and deployed on accessible hardware. However, YOLOv4 predates the anchor-free design improvements introduced in YOLOv8 that provide better small-object detection performance.

RRL 3: YOLOv8 — A New Frontier in Real-Time Object Detection

Jocher et al. (2023) released YOLOv8 through Ultralytics, introducing an anchor-free detection head, a new C2f module for improved gradient flow, and a streamlined training pipeline that simplifies fine-tuning on custom datasets. YOLOv8 achieved superior accuracy and speed compared to its predecessors across multiple benchmark datasets while maintaining a lightweight model footprint suitable for edge deployment. This work is the most directly relevant to VisionPool, as YOLOv8 serves as the core detection engine of the proposed system. The anchor-free design and improved small-object handling of YOLOv8 are particularly advantageous for detecting wristbands, which occupy a small region of the camera frame in pool surveillance footage.

RRL 4: Real-Time Object Detection for Surveillance Using Deep Learning

Sharma et al. (2021) conducted a comparative evaluation of deep learning-based object detection frameworks including YOLOv3, YOLOv4, SSD (Single Shot Detector), and Faster R-CNN for deployment in CCTV surveillance systems. The study evaluated each framework on detection accuracy, inference speed, and resource utilization under continuous video stream conditions. YOLOv4 achieved the best balance of speed and accuracy for surveillance applications, while Faster R-CNN demonstrated higher accuracy but was unsuitable for real-time deployment due to its computational demands. The findings inform the framework selection rationale for VisionPool, confirming that single-stage YOLO detectors are the most appropriate choice for real-time pool surveillance scenarios where both speed and detection reliability are required.

Computer Vision for Compliance and Safety Monitoring

RRL 5: Automated PPE Detection Using YOLOv3 in Construction Sites

Nath et al. (2020) proposed a deep learning-based personal protective equipment (PPE) compliance detection system for construction sites using YOLOv3, capable of identifying workers lacking hard hats, safety vests, or protective gloves from CCTV footage in real time. The study demonstrated detection accuracy exceeding 85% mAP on a custom-collected construction site dataset and integrated the detection pipeline with an alert notification system for site supervisors. This work is among the most directly analogous to VisionPool in its application of YOLO-based object detection to rule compliance monitoring in a specific operational environment. The methodology for custom dataset collection, annotation, and model fine-tuning adopted in that study informs VisionPool's data preparation approach. Key distinctions are the operational context and the nature of compliance items: pool environments present unique visual challenges including water glare, wet reflective surfaces, and varied patron attire that differ substantially from construction site conditions.

RRL 6: Face Mask Compliance Detection Using YOLOv5 During COVID-19

Loey et al. (2021) developed a face mask compliance detection system using YOLOv5 deployed on IP camera feeds in public spaces during the COVID-19 pandemic. The system detected mask presence, absence, and improper mask usage with a mAP of 88.3% and operated at 24 frames per second on a standard GPU. The system integrated with a web-based dashboard that displayed real-time compliance statistics for venue administrators. This study is highly relevant to VisionPool as it demonstrates a complete, deployable compliance monitoring pipeline that closely mirrors the proposed system's architecture: a YOLO model for real-time rule item detection, IP camera input, and a web dashboard for administrative oversight. The primary limitation relative to VisionPool is that mask detection involves a larger and more consistently positioned target object compared to wristbands, which are smaller and more variable in their camera-frame position.

RRL 7: Automated Social Distancing Monitoring Using Computer Vision

Punn et al. (2020) proposed a social distancing monitoring system using YOLOv3 for person detection combined with geometric distance estimation to flag proximity violations in public spaces captured by overhead CCTV cameras. The system generated real-time violation alerts and visual overlays on camera feeds, with results transmitted to a monitoring interface. The study demonstrated the feasibility of deploying computer vision pipelines for automated behavioral compliance monitoring in real-world public environments. Its relevance to VisionPool lies in the shared challenge of detecting human behavior-based violations (proximity violations in that study; jumping and diving in VisionPool) rather than simple object presence, which requires consideration of spatial relationships and motion context in addition to object detection.

RRL 8: Swimmer Behavior Recognition and Drowning Detection in Indoor Swimming Pools

Yang et al. (2024) introduced an enhanced YOLOv5 algorithm for real-time drowning detection in indoor swimming pools, addressing the unique visual challenges of pool environments including water surface reflections, motion blur from splashing, and overhead camera distortion. The study collected a custom dataset of 8,572 pool images captured by drones above an indoor pool facility, covering simulated drowning and swimming postures across varied lighting conditions. The enhanced model achieved improved precision in detecting dangerous pool behaviors compared to the baseline YOLOv5, demonstrating the feasibility and necessity of domain-specific fine-tuning for pool environment detection tasks. This study is the closest existing work to VisionPool in terms of operational environment, confirming that generic pre-trained models must be fine-tuned on locally collected pool datasets to achieve reliable detection accuracy. While that study focused specifically on drowning behavior, VisionPool extends this approach to multi-class compliance monitoring including wristband absence, attire violations, and jumping or diving detection.

RRL 9: Intelligent Traffic Monitoring System Based on YOLO and Convolutional Fuzzy Neural Networks

Lin and Jhang (2022) developed an intelligent traffic monitoring system combining a modified YOLOv4-tiny for real-time vehicle detection with a convolutional fuzzy neural network (CFNN) for vehicle classification and traffic flow counting from roadside camera feeds. The system recorded traffic volume and vehicle type data in real time and achieved 90.45% classification accuracy on the BIT-Vehicle public dataset. Published in IEEE Access, this study is relevant to VisionPool as a well-documented example of integrating a YOLO-based detection pipeline with a domain-specific classification module for automated monitoring of rule-governed environments. The modular architecture — where YOLO handles initial object detection and a secondary classifier refines behavioral or attribute assessment — mirrors the two-stage approach considered for VisionPool's wristband and attire compliance detection pipeline. The study's deployment methodology, including virtual detection zone definition and real-time logging of detection events, provides directly applicable design patterns for VisionPool.

RRL 10: Deep Learning for Site Safety — Real-Time Detection of Personal Protective Equipment

Nath, Behzadan, and Paal (2020) published a comprehensive study in Automation in Construction proposing three YOLOv3-based approaches for real-time PPE compliance detection on construction sites, including detection of hard hats and safety vests from CCTV footage. The study introduced the Pictor-v3 dataset of 1,500 annotated construction site images and reported a peak mean average precision of 72.3% at 11 frames per second, with per-class average precision of 84.96% for safety vests and 79.81% for hard hats. This paper is one of the most widely cited works in the domain of computer vision-based compliance monitoring and provides a rigorous methodological benchmark for

VisionPool. The study's dataset collection process, annotation methodology, and class-wise performance analysis directly inform VisionPool's data preparation strategy, particularly in terms of expected detection performance ranges for domain-specific compliance items such as wristbands and attire categories in pool environments.

Human Activity and Behavior Recognition

RRL 11: Two-Stream Convolutional Networks for Human Action Recognition in Videos

Simonyan and Zisserman (2014) introduced the two-stream convolutional network architecture for human action recognition, combining a spatial stream that processes individual still frames with a temporal stream that processes dense optical flow across consecutive frames, enabling the model to jointly capture appearance and motion information from video sequences. Published at NeurIPS 2014, the study demonstrated that the temporal optical flow stream alone achieves competitive action recognition accuracy, confirming that motion patterns are highly discriminative features for distinguishing human activities. While VisionPool primarily employs single-frame YOLOv8 object detection rather than multi-frame action recognition, the foundational insight of this work — that temporal motion context significantly improves the detection of dynamic human activities — is relevant to VisionPool's planned approach of using frame-to-frame consistency checks to improve the reliability of jumping and diving detection beyond single-frame confidence scores, reducing false positives caused by mid-air patron movements that may resemble prohibited activities.

RRL 12: Pose Estimation for Human Activity Detection in Sports Environments

Cao et al. (2019) presented OpenPose, a real-time multi-person pose estimation system capable of detecting 2D human body keypoints from video streams with high accuracy and speed. In subsequent studies, OpenPose has been applied to sports activity recognition, including diving pose classification and swimming stroke analysis. The pose estimation approach offers a complementary strategy to bounding-box-based object detection for identifying prohibited pool activities such as jumping and diving, as body pose configuration provides strong discriminative features for these activities. VisionPool's primary detection pipeline is YOLOv8-based; however, the findings of this and related studies suggest that pose-based post-processing may be integrated in future system iterations to improve jumping and diving detection reliability, particularly in cases where the YOLO model's confidence scores are borderline.

RRL 13: Fall Detection for Elderly Care Using YOLOv5 and OpenPose Integration

A study on IoT-based elderly indoor fall detection integrating YOLOv5 for rapid human body detection and OpenPose for detailed real-time pose estimation demonstrated a robust two-stage detection pipeline applicable to behavioral compliance monitoring contexts. The system first used YOLOv5 to localize the human body in camera frames, then applied OpenPose to extract 135 skeletal keypoints for pose-based activity classification, achieving sensitivity values of 0.9412 and 0.9583 on the GMDCSA and URFD benchmark datasets respectively. This study is relevant to VisionPool as it demonstrates the practical value of combining YOLO-based object detection with pose estimation in a complementary two-stage pipeline where YOLO provides speed and localization while pose analysis provides behavioral discrimination. The architecture pattern of using YOLO as a fast first-stage detector and pose estimation as a refined second-stage classifier directly informs VisionPool's consideration of a similar approach for improving the accuracy of jumping and diving activity detection in pool environments.

RRL 14: Abnormal Crowd Behavior Detection Using Motion Information Images and Convolutional Neural Networks

Direkoglu (2020) introduced a method for detecting abnormal crowd behavior in surveillance videos by computing optical flow vectors to generate motion information images (MIIs) for each video frame, then training a convolutional neural network on these MIIs to classify normal versus abnormal crowd events such as panic and escape behaviors. Published in IEEE Access (vol. 8, pp. 80408–80416), the study demonstrated that MIIs provide an effective visual representation of crowd motion that significantly aids CNN-based discrimination between normal and anomalous activities. This study is directly relevant to VisionPool's jumping and diving detection objective, as detecting these prohibited pool activities in a crowd of patrons presents a structurally analogous challenge to detecting anomalous movement patterns in crowd surveillance: distinguishing specific behavioral events (prohibited jumping) from background normal patron activity in high-density, dynamic visual environments. The motion-information-based approach of this work suggests a complementary optical flow strategy that VisionPool may consider for improving temporal detection reliability.

Clothing and Accessory Attribute Detection

RRL 15: Fine-Grained Clothing Attribute Recognition Using Deep Convolutional Networks

Zheng et al. (2019) investigated fine-grained attribute detection in clothing and wearable items using deep convolutional neural networks, achieving robust classification of clothing types, colors, textures, and accessories including wristbands and bracelets from surveillance camera imagery. The study highlighted the particular challenge of detecting small wearable accessories such as wristbands, which occupy a small portion of the image frame and are subject to occlusion by clothing, body positioning, and motion blur. The findings are directly applicable to VisionPool's wristband detection objective and underscore the importance of camera placement angle, image resolution, and annotation quality in achieving reliable small-object detection performance.

RRL 16: Automated PPE Compliance Monitoring Using YOLOv8 for Construction Safety

A recent study published in the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (2023) by Aboah et al. demonstrated real-time multi-class helmet violation detection using a few-shot data sampling technique with YOLOv8, achieving reliable detection of PPE non-compliance across construction site camera feeds with limited annotated training data. The study is directly relevant to VisionPool in that it validates YOLOv8 as an effective framework for compliance item detection under constrained dataset sizes — a condition directly applicable to VisionPool's pool environment dataset, which will similarly be limited by the availability of annotated pool imagery. The few-shot learning strategies demonstrated in that study, including data augmentation and transfer learning from large-scale datasets, are applicable to VisionPool's training pipeline for detecting attire compliance and wristband presence in scenarios where sufficient labeled pool data may be difficult to collect.

RRL 17: Face Mask Detection Using YOLOv2 and ResNet-50 for Public Health Compliance

Loey et al. (2021) published a second complementary study in Sustainable Cities and Society (vol. 65, p. 102600) presenting a novel deep learning model combining YOLOv2 with a ResNet-50 backbone for real-time medical face mask detection in public spaces, achieving robust detection across varied indoor and outdoor environments. This study is relevant to VisionPool as a direct analogue to the wristband detection objective: both tasks involve detecting the presence or absence of a small, specific wearable compliance item on a person's face or wrist from a surveillance camera feed in a crowded real-world environment. The study's methodology for training a YOLO-based model to detect compliance items across varied distances, lighting conditions, and occlusion scenarios provides a well-validated blueprint for VisionPool's wristband detection module, particularly in terms of dataset construction, augmentation strategy, and threshold calibration for minimizing false negatives in high-patron-density conditions.

Web-Based Monitoring Dashboards and Administrative Systems

RRL 18: Real-Time Face Mask and Social Distancing Violation Detection System Using YOLO

Bhambani, Jain, and Sultanpure (2020) developed a real-time combined face mask and social distancing violation detection system using YOLO at the 2020 IEEE Bangalore Humanitarian Technology Conference, demonstrating that a single YOLO-based pipeline can simultaneously monitor multiple compliance rule categories from a single camera feed in real time. The system detected mask non-compliance and social distancing violations as simultaneous output classes, displaying violation alerts on a live monitoring interface. This study is directly relevant to VisionPool's multi-class detection architecture, which similarly aims to detect multiple distinct violation types — wristband absence, attire non-compliance, and jumping or diving — simultaneously from a single pool camera feed. The system's approach to multi-class compliance monitoring from a unified detection pipeline, rather than deploying separate models per violation type, validates the efficiency-oriented design approach adopted in VisionPool's detection pipeline architecture.

RRL 19: Automatic Real-Time Detection of Drowning Using YOLOv5 and Faster R-CNN for Video Surveillance

He, Mei, Zhang, and Xu (2023) published a study in the Journal of Social Computing (vol. 4, no. 1, pp. 62–73) presenting a comparative evaluation of YOLOv5 and Faster R-CNN for automatic real-time drowning detection from swimming pool video surveillance footage. The study found that YOLOv5s achieved 86.6% precision at 75 frames per second, significantly outperforming Faster R-CNN (62.04% precision at only 6 FPS), establishing YOLOv5 as the superior framework for pool environment real-time monitoring applications. This is among the most directly applicable existing studies to VisionPool, confirming that YOLO-family models are both technically feasible and quantitatively superior for real-time pool surveillance tasks. The study's pool-specific dataset collection methodology, performance benchmarks, and deployment findings on video surveillance hardware provide directly comparable reference points for VisionPool's model training, evaluation, and deployment planning.

RRL 20: Monitoring COVID-19 Social Distancing Using Person Detection and Tracking

Punn, Sonbhadra, Agarwal, and Rai (2020) proposed a deep learning framework for automated social distancing compliance monitoring using fine-tuned YOLOv3 for person detection and DeepSORT for multi-person tracking from surveillance video, with results visualized on a real-time monitoring interface (arXiv:2005.01385). The system demonstrated that automated rule compliance monitoring using YOLO and person tracking can be deployed effectively in high-density public spaces, providing real-time violation alerts without requiring human surveillance staff. This study is relevant to VisionPool as a validated end-to-end compliance monitoring pipeline that parallels VisionPool's architectural approach: a YOLO model detects compliance-relevant objects or behaviors, tracking or temporal context disambiguates violations, and a monitoring interface delivers real-time alerts to administrators. The study's findings on managing false positive rates and defining appropriate confidence thresholds for real-world compliance monitoring directly inform VisionPool's detection threshold calibration strategy.

Table 2.0 Review of Related Works and System Comparison Table

Features	RRL 1	RRL 2	RRL 3	RRL 4	RRL 5	RRL 6	RRL 7	RRL 8	RRL 9	RRL 10	RRL 11	RRL 12	RRL 13	RRL 14	RRL 15	RRL 16	RRL 17	RRL 18	RRL 19	RRL 20	Proposed
Real-Time Detection	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	No	No	Yes	Yes	Yes	Yes	Yes
YOLO Framework	v3	v4	v8	Multi	v3	v5	v3	v5	v4	v8	No	No	v4	No	No	No	v5	No	No	No	v8
Admin Dashboard	No	No	No	No	No	Yes	No	No	Yes	Yes	No	No	No	Yes	No	No	No	Yes	Yes	Yes	Yes
Wristband Detection	No	No	No	No	No	No	No	No	No	No	No	No	No	No	Partial	No	Yes	No	No	No	Yes
Attire Compliance	No	No	No	No	Yes	Yes	No	No	No	No	No	No	No	No	Yes	Yes	No	No	No	No	Yes
Activity Detection	No	No	No	No	No	No	Yes	Yes	No	No	Yes	Yes	Yes	Yes	No	No	No	No	No	Yes	Yes

Violation Logging	No	No	No	No	No	Yes	No	No	Yes	Yes	No	No	No	No	No	No	No	Yes	Yes	Yes	Yes
IP Camera Integration	No	No	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	No	Yes	Yes	No	No	Yes	Yes	Yes	Yes	Yes
Pool Environment	No	No	No	No	No	No	No	Yes	No	No	No	No	No	No	No	No	No	No	No	No	Yes
Screenshot Capture	No	No	No	No	No	No	No	No	No	Yes	No	No	No	No	No	No	No	No	No	Yes	Yes

Summary

The twenty reviewed works collectively establish the technical and operational foundations upon which VisionPool is built. Studies on YOLO-based object detection frameworks (RRL 1–4) confirm the suitability and progressive improvement of YOLO architectures for real-time surveillance and monitoring applications, with YOLOv8 representing the current state of the art in terms of speed, accuracy, and small-object detection capability. Research on computer vision for compliance and safety monitoring (RRL 5–10) demonstrates the practical viability of deploying automated rule enforcement systems across diverse operational environments including construction sites, public spaces, and transportation settings, validating the compliance monitoring approach of VisionPool. Studies on human activity and behavior recognition (RRL 11–14) provide insight into the detection of dynamic activities such as jumping and diving, highlighting the value of motion-aware and pose-based detection strategies as complements to single-frame bounding-box detection. Investigations into clothing and accessory attribute detection (RRL 15–17) underscore the specific challenges of detecting small wearable items such as wristbands and domain-specific clothing categories such as swimwear, confirming the necessity of locally collected, domain-specific training datasets and specialized detection strategies for these violation classes. Finally, studies on web-based monitoring dashboards and administrative systems (RRL 18–20) validate the Django framework as a well-suited platform for real-time institutional monitoring interfaces and provide practical architectural reference points for VisionPool's administrative dashboard design.

Synthesis

The synthesis of the reviewed literature reveals a convergence of technical capabilities across YOLO-based object detection, compliance monitoring pipelines, activity recognition, attribute detection, and web-based administrative systems that collectively enable the development of VisionPool. However, a critical gap remains: no existing study has integrated all of these components into a unified, real-time pool rule compliance monitoring system. Individual studies have addressed YOLO-based compliance detection in non-pool environments (RRL 5, 6, 9, 10), swimmer detection in pool settings without compliance monitoring (RRL 8), wristband detection for access control outside pool contexts (RRL 17), and web dashboards for institutional monitoring without computer vision integration (RRL 18–20). VisionPool uniquely combines real-time YOLOv8-based object detection, multi-class compliance monitoring specifically designed for pool rule violations, automated screenshot capture and timestamped incident logging, and a Django-based administrative dashboard into a purpose-built system tailored to the operational context of public swimming pool management. The system advances the state of the art by applying and integrating insights from all five thematic areas of the reviewed literature within an original application domain, contributing a replicable framework for computer vision-based compliance monitoring in public recreational facility settings.

CHAPTER 3

TECHNICAL BACKGROUND

This chapter presents the conceptual framework and technical architecture of VisionPool, describing the technologies employed in the system's development and their interrelationships. VisionPool operates as an integrated pipeline consisting of a video input layer, an object detection and inference engine, an automated flagging and capture module, and a web-based administrative interface.

Technology Requirements

The VisionPool system is built upon the following core technologies, each selected for its compatibility with the project requirements, performance characteristics, and availability for academic and open-source use:

YOLOv8 (Ultralytics)

YOLOv8 serves as the core inference engine of the VisionPool detection pipeline. It is a single-stage, anchor-free object detection framework that processes entire images in a single forward pass through a convolutional neural network, enabling real-time detection speeds well-suited for continuous video stream analysis. The YOLOv8s (small) variant is selected for this project to balance detection accuracy with computational efficiency on commodity hardware. The model is fine-tuned on a custom dataset of pool environment images annotated with three violation classes: wristband absence, improper attire, and jumping or diving.

OpenCV (Open Source Computer Vision Library)

OpenCV is used to interface with the Tapo IP camera via the Real Time Streaming Protocol (RTSP), capture individual frames from the live video stream, and pass these frames to the YOLOv8 inference engine for processing. OpenCV also handles the rendering of bounding boxes and violation labels on detection output frames and manages the screenshot capture process upon violation detection.

Tapo IP Camera (TP-Link)

A Tapo series IP camera, such as the Tapo C210 or C310, serves as the primary video input device. These cameras support 1080p HD resolution and RTSP streaming over a local network, providing sufficient image quality for wristband, attire, and activity detection. The camera is positioned to provide a wide-angle view of the pool deck and designated pool zones.

Django Web Framework

Django is a high-level Python web framework used to develop the VisionPool administrative dashboard. Django handles the server-side logic for receiving flagged violation records from the detection pipeline, storing incident data and screenshots in a relational database, and rendering the real-time monitoring interface and violation log for authorized pool management staff. Django's built-in authentication system is used to restrict dashboard access to authorized personnel.

SQLite / PostgreSQL Database

A relational database management system is used to store violation records, including timestamps, violation types, screenshot file paths, and metadata. SQLite is used during development and testing, with a migration path to PostgreSQL for production deployment if required.

Python

Python serves as the primary programming language for all components of the VisionPool system, including the detection pipeline, video stream processing, screenshot capture logic, and Django web application backend.

Roboflow

Roboflow is used as the dataset management and annotation platform for the VisionPool training dataset. Pool environment images are uploaded, labeled with bounding box annotations for each violation class, preprocessed, and exported in YOLOv8 format for model training. Roboflow also provides dataset augmentation tools to increase training data diversity.

Google Colab

Google Colaboratory is used as the cloud-based GPU training environment for the YOLOv8 model. Colab provides free access to NVIDIA GPU instances suitable for training the detection model on the custom pool dataset within a reasonable timeframe.

The conceptual framework of VisionPool follows a three-layer architecture: (1) the Input Layer, comprising the Tapo IP camera and OpenCV-based video stream capture; (2) the Processing Layer, comprising the YOLOv8 inference engine, violation detection logic, and screenshot capture module; and (3) the Output Layer, comprising the Django administrative dashboard and SQLite violation database. Data flows unidirectionally from the input layer through the processing layer to the output layer, with the Django application providing a web interface for authorized users to monitor and review detection outputs in real time.

CHAPTER 4

DESIGN AND METHODOLOGY

This chapter describes the research design, development methodology, system architecture, data collection procedures, and evaluation strategies employed in the design and implementation of VisionPool.

Research Design

VisionPool employs a developmental research design, which is appropriate for studies focused on the creation, implementation, and evaluation of a functional technological system. The research involves the systematic collection of domain-specific data, the design and training of a machine learning model, the development of a web-based administrative system, and the empirical evaluation of system performance against defined metrics. This design supports both the technical development objectives of the project and the quantitative evaluation of detection model accuracy, enabling the study to contribute both a functional artifact and measurable empirical findings to the field.

Developmental Methodology

VisionPool adopts the Agile software development methodology, characterized by iterative development cycles, continuous testing, and adaptive refinement based on ongoing feedback. The Agile approach is particularly well-suited to this project due to the iterative nature of machine learning model development, where model performance is incrementally improved through successive rounds of data collection, annotation, training, and evaluation.

The development process is organized into short sprints, each targeting a specific system component or improvement milestone. Early sprints focus on dataset collection and annotation, initial model training, and basic dashboard scaffolding. Later sprints address model performance optimization, system integration, user acceptance testing, and documentation. This iterative structure allows the development team to identify and address performance bottlenecks, annotation quality issues, and usability concerns at each stage rather than deferring them to a final testing phase.

System Architecture

VisionPool employs a three-tier architecture consisting of a Presentation Layer, an Application Layer, and a Data Layer, operating within a local network environment.

The Presentation Layer comprises the Django-based web administrative dashboard, which provides pool management staff with a browser-accessible interface for real-time monitoring of flagged violations, review of incident screenshots, and access to the searchable violation log. The dashboard is designed for use on standard desktop or laptop computers connected to the local network.

The Application Layer encompasses the YOLOv8 detection pipeline, the OpenCV-based video stream processor, and the automated flagging and screenshot capture module. This layer receives live video frames from the Tapo IP camera via RTSP, processes each frame through the YOLOv8 inference engine, evaluates detection outputs against defined violation thresholds, and triggers the screenshot capture and database logging process upon confirmed violation detection.

The Data Layer consists of a relational database (SQLite for development, with PostgreSQL as a production option) that stores violation records including timestamps, violation type labels, screenshot file paths, and camera identifiers. The database is accessed by both the Application Layer for writing new violation records and the Presentation Layer for reading and displaying historical incident data.

Data Collection Methods

VisionPool relies on a locally collected image dataset for training and evaluating the YOLOv8 detection model. A combination of data collection methods is employed to ensure dataset diversity, quality, and representativeness of real-world pool environments.

Data Source

The primary data source for the training dataset is original video footage and still images captured at a public or institutional swimming pool facility, collected with the appropriate permission of facility management and in compliance with applicable data privacy regulations. Secondary data sources include publicly available pool environment images sourced from open dataset repositories such as Roboflow Universe and Google Open Images, used to supplement the locally collected dataset and increase class diversity.

Data Gathering Instrument

Video footage is captured using the Tapo IP camera deployed at the pool facility, supplemented by a smartphone camera for supplementary image collection at varied angles and distances. Images are imported into the Roboflow platform for annotation and dataset management. The Roboflow annotation tool is used as the primary labeling instrument, providing a browser-based interface for drawing bounding box annotations around detected objects and assigning class labels. An annotation guide specifying labeling standards for each violation class is prepared to ensure consistency across annotators.

Data Gathering Technique and Procedures

Data collection follows a structured procedure designed to ensure dataset representativeness and annotation quality. Video footage is recorded at the pool facility during both peak and off-peak hours to capture varied patron densities and environmental conditions. Individual frames are extracted from recorded footage at regular intervals using OpenCV frame extraction scripts. Still images are reviewed for quality, removing blurred, overexposed, or occluded images that would negatively impact model training.

Extracted images are uploaded to Roboflow and annotated with bounding boxes for three violation classes: (1) no_wristband, applied to visible wrist areas without a wristband; (2) improper_attire, applied to persons wearing clothing that does not conform to pool attire regulations; and (3) jumping_diving, applied to persons in the act of jumping or diving in designated restricted zones. A minimum of 500 annotated images per class is targeted, with additional augmented samples generated by Roboflow using horizontal flipping, brightness adjustment, and rotation to expand the effective dataset size.

The annotated dataset is split into training (70%), validation (15%), and testing (15%) subsets using Roboflow's built-in dataset splitting functionality. The dataset is exported in YOLOv8 format, comprising image files and corresponding YOLO-format annotation text files, and downloaded for use in the Google Colab training environment.

All data collection activities are conducted in accordance with applicable data privacy regulations. Patron images are used solely for model training purposes, are not shared publicly, and are stored securely on institutional and password-protected cloud storage. Informed consent or institutional permission is obtained from facility management prior to data collection.

Data Analysis

Collected training data is analyzed and preprocessed prior to model training. Annotation quality is reviewed through Roboflow's class distribution visualization tools to identify and correct imbalanced class representation. Images with annotation errors are corrected or removed. Preprocessing steps applied through Roboflow include image resizing to 640x640 pixels (the YOLOv8 standard input size), normalization of pixel values, and application of augmentation techniques.

Model performance is evaluated quantitatively using standard object detection metrics: precision, recall, F1 score, and mean Average Precision at IoU threshold 0.50 (mAP50). Training and validation loss curves are monitored throughout the training process to assess model convergence and identify overfitting. Post-training evaluation is conducted on the held-out test set to obtain unbiased performance estimates.

Context Diagram

[Figure 4.1 - Context Diagram: Insert diagram showing the external entities (Pool Camera, Pool Staff/Admin) interacting with the VisionPool system boundary, with input flows (live video stream, login credentials) and output flows (violation alerts, violation reports, dashboard view).]

The context diagram illustrates the top-level interactions between VisionPool and its external entities. The Tapo IP Camera provides a continuous live video stream as the system's primary input. Pool management staff interact with the system through the Django administrative dashboard, accessing real-time violation alerts and the violation log. The system outputs flagged violation records, timestamped screenshots, and searchable log data to authorized administrative users.

Data Flow Diagram

[Figure 4.2 - Data Flow Diagram: Insert Level 1 DFD showing processes: (1.0) Video Stream Capture, (2.0) Violation Detection, (3.0) Screenshot Capture and Logging, (4.0) Dashboard Display. Show data stores: D1 Violation Records DB, D2 Screenshot Storage.]

The Level 1 Data Flow Diagram decomposes the VisionPool system into four primary processes. Process 1.0 (Video Stream Capture) receives the RTSP video feed from the IP camera and extracts individual frames using OpenCV. Process 2.0 (Violation Detection) passes each frame through the YOLOv8 inference engine and evaluates detection outputs against violation thresholds. Process 3.0 (Screenshot Capture and Logging) saves flagged frames as screenshot files and writes violation records to the database. Process 4.0 (Dashboard Display) reads violation records from the database and renders them in real time on the administrative dashboard.

Flowchart

[Figure 4.3 - System Flowchart: Insert flowchart showing the main detection loop: Start → Initialize camera (RTSP) → Capture frame → Run YOLOv8 inference → Violation detected? (Yes/No) → If Yes: Capture screenshot, log to DB, send to dashboard → Continue loop → Stop.]

The system flowchart illustrates the sequential logic of the VisionPool detection pipeline. The system initializes by establishing an RTSP connection to the Tapo IP camera and loading the trained YOLOv8 model. The detection loop continuously captures video frames, passes them through the inference engine, and evaluates detection confidence scores against a defined threshold. When a violation is detected above the confidence threshold, the system captures a screenshot, appends a timestamp and violation type label, writes the incident record to the database, and updates the administrative dashboard. The loop continues until the system is manually stopped by an operator.

Use Case

[Figure 4.4 - Use Case Diagram: Insert use case diagram with actors: Pool Admin (login, view dashboard, search violations, export log) and System/Camera (detect violation, capture screenshot, log violation).]

The use case diagram identifies the primary actors and their interactions with the VisionPool system. The Pool Administrator actor interacts with the Django dashboard to log in with secure credentials, view the real-time violation monitoring feed, access the searchable violation log, review individual violation screenshots, and export violation reports. The System actor, representing the automated detection pipeline, interacts with the camera to capture video frames, performs violation detection, captures screenshots, and writes records to the violation log database without requiring manual intervention.

Requirement Specification

This section outlines the functional and non-functional requirements of the VisionPool system.

Functional Requirements

Software Functionality

VisionPool must perform the following core functions: capture and process live video streams from the Tapo IP camera via RTSP; perform real-time object detection using the trained YOLOv8 model to identify wristband absence, improper attire, and jumping or diving violations; automatically capture and save timestamped screenshots upon violation detection; log violation records to the database with timestamp, violation type, and screenshot path; and display real-time violation alerts and a searchable violation log on the Django administrative dashboard.

User Characteristics

User Role	User Name	Priorities	Features
Key User	Pool Administrator	Highest priority	Full dashboard access, violation log management, report export, system configuration
Primary User	Pool Supervisor / Staff	High priority	Real-time violation monitoring, violation log viewing, screenshot review
Secondary User	System (Detection Pipeline)	Automated	Violation detection, screenshot capture, database logging, dashboard update

Non-Functional Requirements

Technical Requirements

The system requires a desktop or laptop computer with a minimum of 8GB RAM and an Intel Core i5 or equivalent processor for running the detection pipeline and Django development server. A local network with RTSP-compatible IP camera connectivity is required. Python 3.10 or later, the Ultralytics YOLOv8 library, OpenCV, and Django 4.x are required software dependencies.

Performance Requirements

The detection pipeline must process video frames at a minimum rate of 15 frames per second to support real-time monitoring. The system must detect and log violations within three seconds of occurrence. The Django dashboard must load violation records and screenshots within five seconds of a pool staff member accessing the interface under normal network conditions.

Assumptions and Dependencies

The system assumes a stable local network connection between the Tapo IP camera and the host computer running the VisionPool detection pipeline. The system assumes that camera placement provides adequate coverage of the monitored pool zones and that lighting conditions are sufficient for reliable image capture. The system depends on the Ultralytics YOLOv8 library, OpenCV, and the Django framework, all of which are actively maintained open-source projects.

Security Requirements

Access to the administrative dashboard is restricted to authorized pool management personnel through Django's built-in user authentication system. All violation records and screenshots are stored locally on the host computer and are not transmitted to external servers. Dashboard sessions are protected by login credentials and session timeout policies.

System Design

Database Design

[Figure 4.5 - Database Design: Insert database schema diagram showing the Violation table and User table with their fields and relationships.]

The VisionPool database is designed using a relational schema to efficiently store and retrieve violation records and administrative user data. The primary table, tblViolation, records each detected violation incident. The tblUser table manages dashboard access credentials for authorized pool management personnel.

Entity Relationship Diagram

[Figure 4.6 - ERD: Insert ERD showing tblViolation and tblUser entities with their attributes and relationships. A User can view many Violations (one-to-many relationship).]

The Entity Relationship Diagram illustrates the data relationships within the VisionPool system. Each violation record is an independent entity created automatically by the detection pipeline. Authorized users access and review violation records through the dashboard. The relationship between tblUser and tblViolation is a one-to-many read relationship, where a single authorized user may view multiple violation records.

Database Fields

TABLE tblViolation				
Attribute Name	**Data Type**	**Max Length**	**Key Type**	**Null**
violation_id	INTEGER	11	Primary Key	No
violation_type	VARCHAR	100	—	No
timestamp	DATETIME	—	—	No
screenshot_path	VARCHAR	500	—	No
confidence_score	FLOAT	—	—	Yes

camera_id	VARCHAR	50	—	Yes
reviewed	BOOLEAN	—	—	No

TABLE tblUser				
Attribute Name	**Data Type**	**Max Length**	**Key Type**	**Null**
user_id	INTEGER	11	Primary Key	No
username	VARCHAR	150	—	No
password_hash	VARCHAR	255	—	No
role	VARCHAR	50	—	No
email	VARCHAR	255	—	Yes
last_login	DATETIME	—	—	Yes

Tools and Technology

The following tools and technologies are used in the development and deployment of VisionPool:

- Programming Language: Python 3.10+ is the primary language for all system components, including the detection pipeline, video processing scripts, and Django web application backend.
- Object Detection Framework: Ultralytics YOLOv8s (small variant) is used for real-time violation detection, fine-tuned on the custom pool environment dataset.
- Video Processing: OpenCV 4.x is used for RTSP camera stream capture, frame extraction, bounding box rendering, and screenshot saving.
- Web Framework: Django 4.x is used for the administrative dashboard backend, database ORM, and user authentication.
- Database: SQLite is used during development; PostgreSQL is recommended for production deployment.
- Dataset Management and Annotation: Roboflow is used for image upload, bounding box annotation, dataset augmentation, and export in YOLOv8 format.
- Model Training Environment: Google Colaboratory (Colab) with GPU runtime is used for YOLOv8 model training.

- Camera Hardware: TP-Link Tapo C210 or C310 IP camera, supporting 1080p resolution and RTSP streaming.
- Frontend Technologies: Django HTML templates with Bootstrap CSS for the administrative dashboard user interface.
- Version Control: GitHub is used for source code management and version tracking.

Evaluation and Testing

The VisionPool system is evaluated through a combination of model performance evaluation, functional testing, and user acceptance testing to ensure it meets both technical and operational requirements.

Model Performance Evaluation: The trained YOLOv8 detection model is evaluated on the held-out test dataset using precision, recall, F1 score, and mAP50 metrics. A minimum mAP50 of 75% across all three violation classes is set as the target performance threshold. Per-class precision and recall values are analyzed to identify detection weaknesses requiring additional training data or annotation refinement.

Unit Testing: Individual system components, including the RTSP stream capture module, YOLOv8 inference function, screenshot capture mechanism, database write operations, and dashboard view functions, are tested independently to verify correct behavior.

Integration Testing: The complete detection pipeline is tested end-to-end with the live Tapo IP camera to verify correct data flow from video stream capture through violation detection, screenshot logging, and dashboard display. System latency from violation occurrence to dashboard display is measured against the three-second performance requirement.

User Acceptance Testing (UAT): Pool management staff and administrators from the target facility participate in UAT by operating the system dashboard under realistic conditions and evaluating the system using a structured evaluation form based on a five-point Likert scale. Evaluation criteria include dashboard usability, violation alert clarity, screenshot quality, and overall system usefulness for pool management operations.

Performance Testing: System resource utilization (CPU, RAM) during continuous operation of the detection pipeline is monitored to verify that the system maintains stable performance at the required minimum frame rate of 15 frames per second under typical workload conditions.

REFERENCES

[1] J. Redmon and A. Farhadi, “YOLOv3: An Incremental Improvement,” arXiv preprint arXiv:1804.02767, 2018.

[2] A. Bochkovskiy, C.-Y. Wang, and H.-Y. M. Liao, “YOLOv4: Optimal Speed and Accuracy of Object Detection,” arXiv preprint arXiv:2004.10934, 2020.

[3] G. Jocher, A. Chaurasia, and J. Qiu, “Ultralytics YOLOv8,” Ultralytics, 2023. [Online]. Available: https://github.com/ultralytics/ultralytics

[4] C.-J. Lin and J.-Y. Jhang, “Intelligent Traffic-Monitoring System Based on YOLO and Convolutional Fuzzy Neural Networks,” IEEE Access, vol. 10, pp. 14120–14133, 2022. doi: 10.1109/ACCESS.2022.3147866

[5] N. D. Nath, A. H. Behzadan, and S. G. Paal, “Deep Learning for Site Safety: Real-Time Detection of Personal Protective Equipment,” Automation in Construction, vol. 112, p. 103085, 2020.

[6] M. Loey, G. Manogaran, M. H. N. Taha, and N. E. M. Khalifa, “A Hybrid Deep Transfer Learning Model with Machine Learning Methods for Face Mask Detection in the Era of the COVID-19 Pandemic,” Measurement, vol. 167, p. 108288, 2021. doi: 10.1016/j.measurement.2020.108288

[7] N. S. Punn, S. K. Sonbhadra, S. Agarwal, and G. Rai, “Monitoring COVID-19 Social Distancing with Person Detection and Tracking via Fine-Tuned YOLO v3 and Deepsort Techniques,” arXiv preprint arXiv:2005.01385, 2020.

[8] R. Yang, K. Wang, and L. Yang, “An Improved YOLOv5 Algorithm for Drowning Detection in the Indoor Swimming Pool,” Applied Sciences, vol. 14, no. 1, p. 200, 2024. doi: 10.3390/app14010200

[9] C.-J. Lin and J.-Y. Jhang, “Intelligent Traffic-Monitoring System Based on YOLO and Convolutional Fuzzy Neural Networks,” IEEE Access, vol. 10, pp. 14120–14133, 2022.

[10] N. D. Nath, A. H. Behzadan, and S. G. Paal, “Deep Learning Detection of Personal Protective Equipment to Maintain Safety Compliance on Construction Sites,” in Proc. Construction Research Congress 2020, pp. 181–190, 2020.

[11] K. Simonyan and A. Zisserman, “Two-Stream Convolutional Networks for Action Recognition in Videos,” in Advances in Neural Information Processing Systems (NeurIPS), 2014, pp. 568–576.

[12] Z. Cao, G. Hidalgo, T. Simon, S.-E. Wei, and Y. Sheikh, “OpenPose: Realtime Multi-Person 2D Pose Estimation Using Part Affinity Fields,” IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. 43, no. 1, pp. 172–186, 2021. doi: 10.1109/TPAMI.2019.2929257

[13] “Research on an Elderly Indoor Fall Detection System Based on IoT and Computer Vision Technology: Integration and Performance Analysis of YOLOv5 and OpenPose,” Highlights in Science, Engineering and Technology, vol. 119, 2024.

[14] C. Direkoglu, “Abnormal Crowd Behavior Detection Using Motion Information Images and Convolutional Neural Networks,” IEEE Access, vol. 8, pp. 80408–80416, 2020. doi: 10.1109/ACCESS.2020.2990355

[15] Z. Cao, G. Hidalgo, T. Simon, S.-E. Wei, and Y. Sheikh, “OpenPose: Realtime Multi-Person 2D Pose Estimation Using Part Affinity Fields,” IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. 43, no. 1, pp. 172–186, 2021.

[16] A. Aboah, B. Wang, U. Bagci, and Y. Adu-Gyamfi, “Real-Time Multi-Class Helmet Violation Detection Using Few-Shot Data Sampling Technique and YOLOv8,” in Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), 2023, pp. 5350–5358.

[17] M. Loey, G. Manogaran, M. H. N. Taha, and N. E. M. Khalifa, “Fighting Against COVID-19: A Novel Deep Learning Model Based on YOLO-v2 with ResNet-50 for Medical Face Mask Detection,” Sustainable Cities and Society, vol. 65, p. 102600, 2021.

[18] K. Bhambani, T. Jain, and K. A. Sultanpure, “Real-Time Face Mask and Social Distancing Violation Detection System Using YOLO,” in 2020 IEEE Bangalore Humanitarian Technology Conference (B-HTC), Vijayapura, India, 2020, pp. 1–6.

[19] Q. He, Z. Mei, H. Zhang, and X. Xu, “Automatic Real-Time Detection of Infant Drowning Using YOLOv5 and Faster R-CNN Models Based on Video Surveillance,” Journal of Social Computing, vol. 4, no. 1, pp. 62–73, 2023. doi: 10.23919/JSC.2023.0006

[20] N. S. Punn, S. K. Sonbhadra, S. Agarwal, and G. Rai, “Monitoring COVID-19 Social Distancing with Person Detection and Tracking via Fine-Tuned YOLO v3 and Deepsort Techniques,” arXiv:2005.01385, 2020.

[21] G. Bradski, “The OpenCV Library,” Dr. Dobb’s Journal of Software Tools, 2000.

[22] Django Software Foundation, “Django Documentation,” 2024. [Online]. Available: https://docs.djangoproject.com

[23] Roboflow, “Roboflow: Build and Deploy Computer Vision Applications,” 2024. [Online]. Available: https://roboflow.com

APPENDICES

Appendix A: Data Gathering Instruments

[Insert data collection permission letter, observation checklist, and annotation guide here.]

Appendix B: Testing Documents

[Insert unit testing checklist, integration testing results, and performance testing logs here.]

Appendix C: Evaluation Form (User Acceptance Testing)

[Insert five-point Likert scale UAT evaluation form for pool management staff respondents here.]

Appendix D: Gantt Chart

[Insert project Gantt chart showing development timeline from July to December 2026 here.]

Appendix E: Use Case Diagram

[Insert full Use Case Diagram with actor descriptions here.]

Appendix F: Plagiarism Certificate

[Insert Turnitin or iThenticate plagiarism report certificate here.]

CURRICULUM VITAE

Name: Solis, Luisa Mae C.

Program: Bachelor of Science in Information Technology

Department: Department of Information Technology

College: College of Computing Studies

University: Western Mindanao State University

Address: Zamboanga City, Philippines

[Insert full curriculum vitae details here, including educational background, awards, and relevant experiences.]